Further Strategies — problem 14.69

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For every pair of real numbers a and b, we consider the sequence $p_n = \lfloor 2\{an + b\} \rfloor$. Here $\{c\}$ is the fractional part of c. We call any k successive terms of this sequence a *word*. Is it true that any sequence of zeros and ones of length k is a word of a sequence given by some a and b : (a) for $k = 4$; (b) for $k = 5$ (MMO 1993) ?